

PROFESSIONAL EXPERIENCE

California Institute of Technology, USA

Postdoctoral Research Scholar Associate

Sep 2023–Aug 2026

- Working within the LIGO Laboratory and Theoretical Astrophysics including Relativity (TAPIR) Research Group

EDUCATION

University of Birmingham, Institute for Gravitational Wave Astronomy, UK

Ph.D. *Physics and Astronomy*, supervised by Dr. Patricia Schmidt

Sep 2019–Aug 2023

- Thesis title: “*Gravitational Waveform Modelling for Inference of Spin Precession in Binary Black Hole Mergers*” - [read the thesis](#)
- Funded by the University of Birmingham and Science and Technologies Funding Council, UK
- PhD Award Date (graduation): 6th Dec 2023

King’s College London, UK

MSc. *Physics*, thesis supervised by Prof. Eugene Lim

Sep 2018–Aug 2019

- Thesis title: “*Modelling Critical Phenomena in Scalar Field Collapse*”

Grad. Dip. *Physics*, thesis supervised by Prof. Mairi Sakellariadou

Sep 2017–Aug 2018

- Thesis title: “*Stochastic Background of Gravitational Waves from Early Universe Phase Transition Bubble Collisions*”

University of Cambridge, UK

M.A. *Hons (Cantab) Natural Sciences*

Oct 2013–Jul 2017

- Part I Mathematics, Part II Astrophysics
- Project title: “*Chaos in the Encounter Map of the Three-Body Problem*”

AWARDS, PRIZES AND FUNDING

- **Marie Skłodowska-Curie Postdoctoral Fellowship (MSCA-PF)**
Gravitational Waveform Uncertainty in Modelling with Bayesian Machine-Learning Enhanced Surrogates (GWUMBLES)
Funded by European Commission/Horizon Europe, hosted by University of Glasgow, UK
Expected dates Oct 2026 - Sep 2028
Awarded with a score of 98.0/100
- **Grant Proposal Contributor** NSF Award Number 2207758
“*Gravitational Wave Physics and Astrophysics with LIGO*”, PI: Prof. Alan J. Weinstein.
Significant written contribution to the awarded grant, roughly 1/3 of the research proposal content
- **Winner 2022** BritGrav22 Best Student Talk
“*A Precessing, Higher-Mode Surrogate Waveform Model Using Neural Networks*” - [watch the video](#)
- **Winner 2022** University of Birmingham School of Physics Student Paper of the Year Award
“*New Effective Precession Spin for Modeling Multimodal Gravitational Waveforms in the Strong-Field Regime*”
- **Overall Winner 2021** University of Birmingham Research Poster Conference Competition
“*You Spin Me Right Round: Modelling ‘Wobbly’ Gravitational Waves from Binary Black Holes*” - [view the poster](#)
- **Winner 2020** UK 3 Minute Thesis Competition
“*Gravity is Pathetic! Shedding Light on Gravity with Models of Spinning Black Hole Mergers*” - [watch the video](#)

POSITIONS OF SCIENTIFIC LEADERSHIP AND SERVICE

- **Co-chair of LIGO-Virgo-KAGRA Waveforms working group** Feb 2025–Current
 - Strategic leadership for waveform development, review and adoption in the collaboration
 - Oversee internal reviews of waveform models before they are integrated into analysis pipelines
 - Represent the Waveforms group interests to the collaboration’s executive bodies
- **Chair of Simulating eXtreme Spacetimes Surrogates working group calls** Jan 2025–Current
 - Organising and coordinating regular meetings
- **Caltech LIGO Undergraduate Study Group Organiser** Oct 2023–Current
 - Preparing and delivering introductory seminar sessions and tutorials on gravitational wave data analysis for interested undergraduate students
- **Caltech LIGO Seminar Series Organiser** Sep 2023–Current
- **LIGO-Virgo-KAGRA Parameter Estimation Expert** Sep 2019–Current
 - Senior expert member for online gravitational wave event parameter estimation rota
 - Responsible for running offline production parameter estimation
 - Responsible for online and offline results review
- **Postgraduate Student Representative, Astrophysics and Space Research Group** Oct 2019–Aug 2023

INVITED CONFERENCE TALKS

- **University of Illinois, Urbana-Champaign, Invited Talk** 1st Jul 2025
“*2G+XG Science*”
As part of the *Physics at the Xtreme (PAX)-X and Cosmic Explorer Symposium*.
- **University of Illinois, Urbana-Champaign, Invited Panel Discussion** 30th Jun 2025
“*Waveforms: Status, Challenges and Opportunities*”
As part of the *Physics at the Xtreme (PAX)-X and Cosmic Explorer Symposium*.
- **Institute for Computational and Experimental Research in Mathematics, Brown University, Invited Talk** 3rd Jun 2025
“Public Code Packages to Visualise and Optimise Gravitational Wave Surrogate Models”
As part of the workshop ‘*Scientific Machine Learning for Gravitational Wave Astronomy*’.
- **APS Global Physics Summit, Anaheim, Invited Talk** 18th Mar 2025
“Building Binary Black Hole Waveform Models Using Numerical Relativity”
As part of the special session ‘*20 Years Since of Numerical Relativity*’.
- **Royal Astronomical Society, London, Invited Talk and Panel Discussion** 10th Jan 2025
“Machine Learning in Parameter Estimation”
As part of the special discussion workshop ‘*Gravitational Wave Data Analysis in the Era of Machine Learning*’.
- **Institute for Pure and Applied Mathematics, UCLA, Invited Talk** 14th Nov 2021
“A New Effective Precession Spin for the Strong-Field Regime”
Given as invited core participant of the long program “*Mathematical and Computational Challenges in the Era of Gravitational Wave Astronomy*”.

CONTRIBUTED CONFERENCE TALKS

- **APS Global Physics Summit, Denver, Contributed Talk** 17th Mar 2026
“*Revisiting the Coprecessing Frame in the Presence of Orbital Eccentricity*”
- **APS Global Physics Summit, Anaheim, Contributed Talk** 18th Mar 2025
“*Optimizing Neural Network Surrogate Models: Application to Black Hole Merger Remnants*”
- **APS April Meeting, Sacramento, Contributed Talk** 5th Apr 2024
“*Constraining Precession in the Black Hole Binary Population with Next-Generation Ground-Based Gravitational Wave Detectors*”
- **BritGrav22, Contributed Talk (virtual)** 4th Apr 2022

“A Precessing, Higher-Mode Surrogate Waveform Model Using Neural Networks”
Winner of Best Student Talk

- **APS April Meeting**, Contributed Talk (virtual) 18th Apr 2021
“Modelling Precessing Higher Modes in the Strong-Field Regime”
- **BritGrav21**, Contributed Talk (virtual) 12th Apr 2021
“Modelling Precessing Higher Modes in the Strong-Field Regime”

INVITED SEMINARS

- **LIGO-Virgo-KAGRA Public GWTC-4.0 Methods Webinar Presenter and Panelist** 19th Feb 2026
“GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients” - [watch the recording](#)
Presenter and panelist for parameter estimation methods
- **LIGO-Virgo-KAGRA Public GWTC-4.0 Results Webinar Presenter and Panelist** 12th Feb 2026
“GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from O4a” - [watch the recording](#)
Presenter and panelist for the source properties results
- **University of Southampton, Southampton** 9th Jan 2025
“Machine Learning Applications to Gravitational Wave Modelling”
Southampton Theory Astrophysics and Gravity (STAG) Seminar Series
- **University of Birmingham, Birmingham** 14th Dec 2023
‘Inferring the Spins of the Black Hole Binary Population with Next-Generation Ground-Based Gravitational Wave Detectors’
Astrophysics Seminar Series
- **University of Glasgow, Glasgow** 1st Dec 2023
“SEOBNN_v4PHM_4dq2: Faster Precessing, Higher Multipole Waveforms with Neural Networks”
Given at the Institute for Gravitational Research (IGR) Data Analysis Group Meeting
- **California Institute of Technology, Pasadena** 10th Nov 2021
“A New Effective Precession Spin for the Strong-Field Regime”
Given at the Caltech LIGO Laboratory Data Analysis Group Meeting

TEACHING AND MENTORING EXPERIENCE

- **Undergraduate Research Internship Mentor (START)**, Caltech 25th Aug 2025–Current
Mentoring community college undergraduate student for a year-long part-time internship project as part of a national NSF program to match community college students with research mentors. The project explores the distinguishability of dynamically-formed subpopulations in a population of binary black holes detected through gravitational waves.
- **Undergraduate Summer Student Mentor (LIGO SURF)**, Caltech 17th Jun 2025–22nd Aug 2025
Mentored one undergraduate student for a 10-week project assessing modelling assumptions for eccentric and precessing gravitational waveforms. This project resulted in a short author publication [arXiv:2603.29307 \[gr-qc\]](#).
- **Undergraduate Summer Student Mentor (LIGO SURF)**, Caltech 18th Jun 2024–23rd Aug 2024
Mentored an undergraduate student 10-week summer project on eccentric and precessing inspiral gravitational waveform model generation. Co-mentored 2 other undergraduate summer students on projects on symbolic regression for trends in gravitational wave data, and statistical methods for assessing the impact of glitches on parameter inference.
- **Undergraduate Summer Student Lecturer**, Caltech 18th Jun 2024–23rd Aug 2024
Gave two interactive lectures for undergraduate summer students entitled *“Using Git for Research”* and *“Waveform Modelling for Compact Binary Coalescences”*.
- **Graduate Student Demonstrator**, University of Birmingham 1st Sep 2020–30th Jul 2023
Helped teach and assess numerous undergraduate courses in Python computing, mathematics for first year undergraduate physicists, and physics labs.

PUBLICATIONS

Publications can also be viewed via [ORCID](#), [INSPIRE](#) or on [arXiv](#).

Short-Author Publications

1. **L. M. Thomas**, K. Chatziioannou, S. Johar, T. Knapp, M. Boyle, “*Revisiting the Coprecessing Frame in the Presence of Orbital Eccentricity*”, submitted to *Phys. Rev. D* April 2026, [arXiv:2603.29307 \[gr-qc\]](#)
2. **L. M. Thomas**, K. Chatziioannou, V. Varma, S. Field, “*Optimizing neural network surrogate models: Application to black hole merger remnants*”, *Phys. Rev. D* **111**, 104029 (2025), [arXiv:2501.16462 \[gr-qc\]](#)
3. **L. M. Thomas**, G. Pratten, P. Schmidt, “*Accelerating multimodal gravitational waveforms from precessing compact binaries with artificial neural networks*”, *Phys. Rev. D* **106**, 104029 (2022), [arXiv:2205.14066 \[gr-qc\]](#)
4. **L. M. Thomas**, P. Schmidt, G. Pratten, “*New effective precession spin for modeling multimodal gravitational waveforms in the strong-field regime*”, *Phys. Rev. D* **103**, 083022 (2021), [arXiv:2012.02209 \[gr-qc\]](#)
5. P. Retegno, G. Pratten, **L. M. Thomas**, P. Schmidt, T. Damour, “*Strong-field scattering of two spinning black holes: Numerical relativity versus post-Minkowskian gravity*”, *Phys. Rev. D* **108**, 124016 (2023), [arXiv:2307.06999 \[gr-qc\]](#)
6. A. Klein, et al. (13 authors incl. **L. M. Thomas**), “*The last three years: multiband gravitational-wave observations of stellar-mass binary black holes*”, 2022, [arXiv:2204.03423 \[astro-ph.HE\]](#)
7. D. Gerosa, M. Mould, D. Gangardt, P. Schmidt, G. Pratten, **L. M. Thomas**, “*A generalized precession parameter χ_p to interpret gravitational-wave data*”, *Phys. Rev. D* **103**, 064067 (2021), [arXiv:2011.11948 \[gr-qc\]](#)
8. G. Pratten, P. Schmidt, R. Buscicchio, **L. M. Thomas**, “*Measuring precession in asymmetric compact binaries*”, *Phys. Rev. Res.* **2**, 043096 (2020), [arXiv:2006.16153 \[gr-qc\]](#)

LIGO-Virgo-KAGRA Collaboration Publications with Direct Contributions

9. LIGO Scientific and Virgo and KAGRA Collaborations (1779 authors incl. **L. M. Thomas**), “*GWTC-4.0: Tests of General Relativity. III. Tests of the Remnants*”, submitted to *Astrophys. J.* March 2026, [arXiv:2603.19021 \[gr-qc\]](#)
 - Expert parameter estimation analyst responsible for execution and validation of inference runs
 - Reviewer of parameter estimation results
10. LIGO Scientific and Virgo and KAGRA Collaborations (1783 authors incl. **L. M. Thomas**), “*GWTC-4.0: Tests of General Relativity. II. Parameterized Tests*”, submitted to *Astrophys. J.* March 2026, [arXiv:2603.19020 \[gr-qc\]](#)
 - Expert parameter estimation analyst responsible for execution and validation of inference runs
 - Reviewer of parameter estimation results
11. LIGO Scientific and Virgo and KAGRA Collaborations (1781 authors incl. **L. M. Thomas**), “*GWTC-4.0: Tests of General Relativity. I. Overview and General Tests*”, submitted to *Astrophys. J.* March 2026, [arXiv:2603.19019 \[gr-qc\]](#)
 - Expert parameter estimation analyst responsible for execution and validation of inference runs
 - Reviewer of parameter estimation results
12. LIGO Scientific and Virgo and KAGRA Collaborations (1809 authors incl. **L. M. Thomas**), “*GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients*”, accepted for publication in *Astrophys. J.* February 2026, [arXiv:2508.18081 \[gr-qc\]](#)
 - Co-lead of paper writing team for parameter estimation
13. LIGO Scientific and Virgo and KAGRA Collaborations (1766 authors incl. **L. M. Thomas**), “*GWTC-4.0: Searches for Gravitational-Wave Lensing Signatures*”, submitted to *Astrophys. J.* February 2026, [arXiv:2512.16347 \[gr-qc\]](#)
 - Expert parameter estimation analyst responsible for execution and validation of inference runs
 - Reviewer of parameter estimation results
14. LIGO Scientific and Virgo and KAGRA Collaborations (1771 authors incl. **L. M. Thomas**), “*GW231123: A Binary Black Hole Merger with Total Mass 190–265 $M_\odot$* ”, *Astrophys. J. Lett.* **993**, L25 (2025), [arXiv:2507.08219 \[astro-ph.HE\]](#)
 - Expert parameter estimation analyst responsible for execution and validation of inference runs
 - Reviewer of parameter estimation results
 - Reviewer of numerical relativity injection infrastructure and results for systematic injection-and-recovery parameter estimation analyses

15. LIGO Scientific and Virgo and KAGRA Collaborations (1768 authors incl. **L. M. Thomas**), “*Open Data from LIGO, Virgo, and KAGRA through the First Part of the Fourth Observing Run*”, accepted for publication in *Astrophys. J.* November 2025, [arXiv:2508.18079 \[gr-qc\]](#)
 - Expert parameter estimation analyst responsible for execution and validation of inference runs
 - Reviewer of parameter estimation results
 - Reviewer of Zenodo public data release
16. LIGO Scientific and Virgo and KAGRA Collaborations (1783 authors incl. **L. M. Thomas**), “*GW241011 and GW241110: Exploring Binary Formation and Fundamental Physics with Asymmetric, High-spin Black Hole Coalescences*”, *Astrophys. J. Lett.* **993**, L21 (2025), [arXiv:2510.26931 \[astro-ph.HE\]](#)
 - Reviewer of paper figures and macros generation for eccentric and quasi-circular parameter estimation results
 - Paper writing review contributor
17. LIGO Scientific and Virgo and KAGRA Collaborations (1772 authors incl. **L. M. Thomas**), “*GWTC-4.0: Constraints on the Cosmic Expansion Rate and Modified Gravitational-wave Propagation*”, submitted to *Astrophys. J.* September 2025, [arXiv:2509.04348 \[astro-ph.CO\]](#)
 - Expert parameter estimation analyst responsible for execution and validation of inference runs
 - Reviewer of parameter estimation results
18. LIGO Scientific and Virgo and KAGRA Collaborations (1805 authors incl. **L. M. Thomas**), “*GWTC-4.0: Population Properties of Merging Compact Binaries*”, submitted to *Astrophys. J.* August 2025, [arXiv:2508.18083 \[astro-ph.HE\]](#)
 - Expert parameter estimation analyst responsible for execution and validation of inference runs
 - Reviewer of parameter estimation results
19. LIGO Scientific and Virgo and KAGRA Collaborations (1345 authors incl. **L. M. Thomas**), “*Tests of general relativity with binary black holes from the second LIGO-Virgo gravitational-wave transient catalog*”, *Phys. Rev. D* **103**, 122002 (2021), [arXiv:2010.14529 \[gr-qc\]](#)
 - Parameter estimation analyst responsible for execution of inference runs
20. LIGO Scientific and Virgo and KAGRA Collaborations (1785 authors incl. **L. M. Thomas**), “*GW250114: Testing Hawking’s Area Law and the Kerr Nature of Black Holes*”, *Phys. Rev. Lett.* **135**, 111403 (2025), [arXiv:2509.08054 \[gr-qc\]](#)
 - Review contributor for parameter estimation results
21. LIGO Scientific and Virgo and KAGRA Collaborations (incl. **L. M. Thomas**), “*GW230814: investigation of a loud gravitational-wave signal observed with a single detector*”, accepted for publication in *Astrophys. J.* November 2025, [arXiv:2509.07348 \[gr-qc\]](#)
 - Member of science case study team for event, with expertise in parameter estimation and waveform systematics to assess suitability of the event for publication in a special event paper
 - Expert parameter estimation analyst responsible for execution and validation of inference runs
22. LIGO Scientific and Virgo and KAGRA Collaborations (1770 authors incl. **L. M. Thomas**), “*GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run*”, accepted for publication in *Astrophys. J.* February 2026, [arXiv:2508.18082 \[gr-qc\]](#)
 - Expert parameter estimation analyst responsible for execution and validation of inference runs
 - Reviewer of parameter estimation results
 - Lead of paper writing team for the ‘source properties’ section of the article
 - Lead producer of figures and tables for the ‘source properties’ section of the article
23. LIGO Scientific and Virgo and KAGRA Collaborations (1659 authors incl. **L. M. Thomas**), “*GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo during the Second Part of the Third Observing Run*”, *Phys. Rev. X* **13**, 041039 (2023), [arXiv:2111.03606 \[gr-qc\]](#)
 - Parameter estimation analyst responsible for execution of inference runs
24. LIGO Scientific and Virgo and KAGRA Collaborations (1741 authors incl. **L. M. Thomas**), “*Open Data from the Third Observing Run of LIGO, Virgo, KAGRA, and GEO*”, *Astrophys. J. Suppl.* **267**, 29 (2023), [arXiv:2302.03676 \[gr-qc\]](#)

- Parameter estimation analyst responsible for execution of inference runs
25. LIGO Scientific and Virgo and KAGRA Collaborations (1430 authors incl. **L. M. Thomas**), “*GWTC-2.1: Deep extended catalog of compact binary coalescences observed by LIGO and Virgo during the first half of the third observing run*”, *Phys. Rev. D* **109**, 022001 (2024), [arXiv:2108.01045 \[gr-qc\]](#)
 - Parameter estimation analyst responsible for execution of inference runs
 26. LIGO Scientific and Virgo and KAGRA Collaborations (1352 authors incl. **L. M. Thomas**), “*GWTC-2: Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run*”, *Phys. Rev. X* **11**, 021053 (2021), [arXiv:2010.14527 \[gr-qc\]](#)
 - Parameter estimation analyst responsible for execution of inference runs
 27. LIGO Scientific and Virgo and KAGRA Collaborations (1339 authors incl. **L. M. Thomas**), “*Population Properties of Compact Objects from the Second LIGO-Virgo Gravitational-Wave Transient Catalog*”, *Astrophys. J. Lett.* **913**, L7 (2021), [arXiv:2010.14533 \[astro-ph.HE\]](#)
 - Parameter estimation analyst responsible for execution of inference runs

Other LIGO-Virgo-KAGRA Collaboration Publications

28. LIGO Scientific and Virgo and KAGRA Collaborations (1764 authors incl. **L. M. Thomas**), “*Searches for Continuous Gravitational Waves from Supernova Remnants in the first part of the LIGO-Virgo-KAGRA Fourth Observing run*”, 2026, [arXiv:2603.25808 \[gr-qc\]](#)
29. LIGO Scientific and Virgo and KAGRA Collaborations (1853 authors incl. **L. M. Thomas**), “*Narrowband searches for continuous gravitational waves from known pulsars in the first two parts of the fourth LIGO-Virgo-KAGRA observing run*”, 2026, [arXiv:2603.25938 \[gr-qc\]](#)
30. LIGO Scientific and Virgo and KAGRA Collaborations (1826 authors incl. **L. M. Thomas**), “*All-sky Searches for Continuous Gravitational Waves from Isolated Neutron Stars in the Data from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run*”, 2026, [arXiv:2603.14168 \[gr-qc\]](#)
31. IceCube, LIGO Scientific and Virgo and KAGRA Collaborations (2214 authors incl. **L. M. Thomas**), “*Deep Search for Joint Sources of Gravitational Waves and High-Energy Neutrinos with IceCube During the Third Observing Run of LIGO and Virgo*”, 2026, [arXiv:2601.07595 \[astro-ph.HE\]](#)
32. LIGO Scientific and Virgo and KAGRA Collaborations (1774 authors incl. **L. M. Thomas**), “*Constraints on gravitational waves from the 2024 Vela pulsar glitch*”, 2026, [arXiv:2512.17990 \[gr-qc\]](#)
33. LIGO Scientific and Virgo and KAGRA Collaborations (1767 authors incl. **L. M. Thomas**), “*Direct multi-model dark-matter search with gravitational-wave interferometers using data from the first part of the fourth LIGO-Virgo-KAGRA observing run*”, 2025, [arXiv:2510.27022 \[astro-ph.CO\]](#)
34. LIGO Scientific and Virgo and KAGRA Collaborations (1765 authors incl. **L. M. Thomas**), “*Search for planetary-mass ultra-compact binaries using data from the first part of the LIGO-Virgo-KAGRA fourth observing run*”, 2025, [arXiv:2511.19911 \[gr-qc\]](#)
35. LIGO Scientific and Virgo and KAGRA Collaborations (1765 authors incl. **L. M. Thomas**), “*All-sky search for continuous gravitational-wave signals from unknown neutron stars in binary systems in the first part of the fourth LIGO-Virgo-KAGRA observing run*”, 2025, [arXiv:2511.16863 \[gr-qc\]](#)
36. LIGO Scientific and Virgo and KAGRA Collaborations (1769 authors incl. **L. M. Thomas**), “*Cosmological and High Energy Physics implications from gravitational-wave background searches in LIGO-Virgo-KAGRA’s O1-O4a runs*”, 2025, [arXiv:2510.26848 \[gr-qc\]](#)
37. LIGO Scientific and Virgo and KAGRA Collaborations (1765 authors incl. **L. M. Thomas**), “*Directional Search for Persistent Gravitational Waves: Results from the First Part of LIGO-Virgo-KAGRA’s Fourth Observing Run*”, 2025, [arXiv:2510.17487 \[gr-qc\]](#)
38. LIGO Scientific and Virgo and KAGRA Collaborations (1816 authors incl. **L. M. Thomas**), “*Search for Continuous Gravitational Waves from Known Pulsars in the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run*”, *Astrophys. J.* **983**, 99 (2025), [arXiv:2501.01495 \[astro-ph.HE\]](#)
39. LIGO Scientific and Virgo and KAGRA Collaborations (1808 authors incl. **L. M. Thomas**), “*GWTC-4.0: An Introduction to Version 4.0 of the Gravitational-Wave Transient Catalog*”, *Astrophys. J. Lett.* **995**, L18 (2025), [arXiv:2508.18080 \[gr-qc\]](#)

40. LIGO Scientific and Virgo and KAGRA Collaborations (1769 authors incl. **L. M. Thomas**), “*Directed searches for gravitational waves from ultralight vector boson clouds around merger remnant and galactic black holes during the first part of the fourth LIGO-Virgo-KAGRA observing run*”, 2025, [arXiv:2509.07352 \[gr-qc\]](#)
41. LIGO Scientific and Virgo and KAGRA Collaborations (incl. **L. M. Thomas**), “*Black Hole Spectroscopy and Tests of General Relativity with GW250114*”, *Phys. Rev. Lett.* **136**, 041403 (2026), [arXiv:2509.08099 \[gr-qc\]](#)
42. LIGO Scientific and Virgo and KAGRA Collaborations (1773 authors incl. **L. M. Thomas**), “*Upper Limits on the Isotropic Gravitational-Wave Background from the first part of LIGO, Virgo, and KAGRA’s fourth Observing Run*”, 2025, [arXiv:2508.20721 \[gr-qc\]](#)
43. LIGO Scientific and Virgo and KAGRA Collaborations (1772 authors incl. **L. M. Thomas**), “*All-sky search for long-duration gravitational-wave transients in the first part of the fourth LIGO-Virgo-KAGRA Observing run*”, 2025, [arXiv:2507.12282 \[gr-qc\]](#)
44. LIGO Scientific and Virgo and KAGRA Collaborations (1780 authors incl. **L. M. Thomas**), “*A Search Using GEO600 for Gravitational Waves Coincident with Fast Radio Bursts from SGR 1935+2154*”, *Astrophys. J.* **977**, 255 (2024), [arXiv:2410.09151 \[astro-ph.HE\]](#)
45. LIGO Scientific and Virgo and KAGRA Collaborations (1822 authors incl. **L. M. Thomas**), “*Swift-BAT GUANO Follow-up of Gravitational-wave Triggers in the Third LIGO–Virgo–KAGRA Observing Run*”, *Astrophys. J.* **980**, 207 (2025), [arXiv:2407.12867 \[astro-ph.HE\]](#)
46. LIGO Scientific and Virgo and KAGRA Collaborations (1780 authors incl. **L. M. Thomas**), “*Search for Gravitational Waves Emitted from SN 2023ixf*”, *Astrophys. J.* **985**, 183 (2025), [arXiv:2410.16565 \[astro-ph.HE\]](#)
47. LIGO Scientific and Virgo and KAGRA Collaborations (1793 authors incl. **L. M. Thomas**), “*Observation of Gravitational Waves from the Coalescence of a 2.5–4.5 $M_{\odot}$ Compact Object and a Neutron Star*”, *Astrophys. J. Lett.* **970**, L34 (2024), [arXiv:2404.04248 \[astro-ph.HE\]](#)
48. LIGO Scientific and Virgo and KAGRA Collaborations (1800 authors incl. **L. M. Thomas**), “*Ultralight vector dark matter search using data from the KAGRA O3GK run*”, *Phys. Rev. D* **110**, 042001 (2024), [arXiv:2403.03004 \[astro-ph.CO\]](#)
49. LIGO Scientific and Virgo and KAGRA Collaborations (1772 authors incl. **L. M. Thomas**), “*Search for Eccentric Black Hole Coalescences during the Third Observing Run of LIGO and Virgo*”, *Astrophys. J.* **973**, 132 (2024), [arXiv:2308.03822 \[astro-ph.HE\]](#)